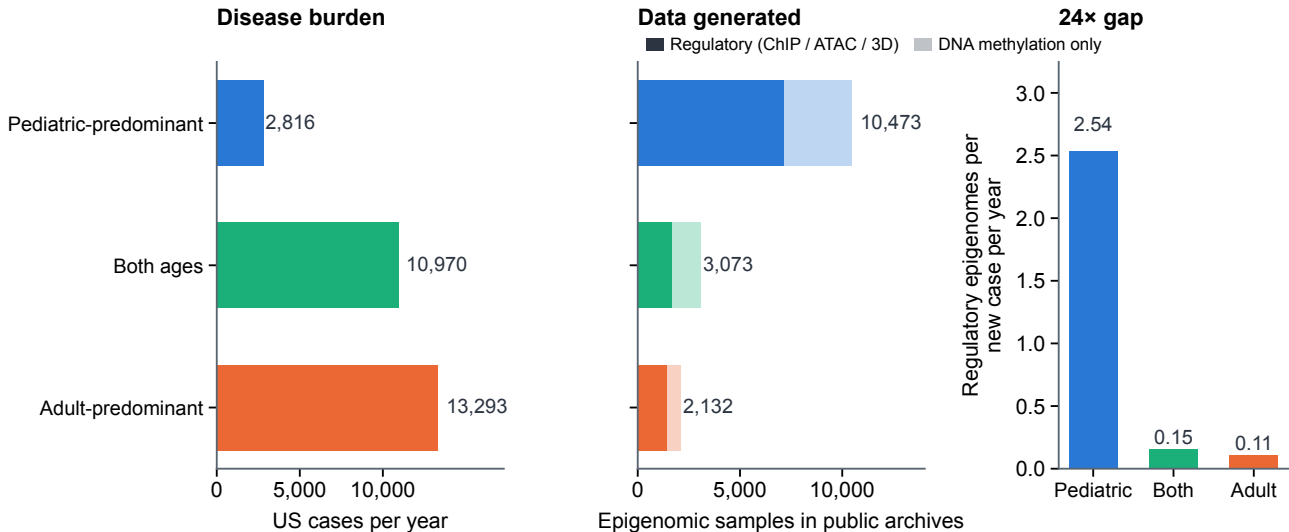


Adult sarcomas carry ~5× the disease burden and a small fraction of the epigenomic data

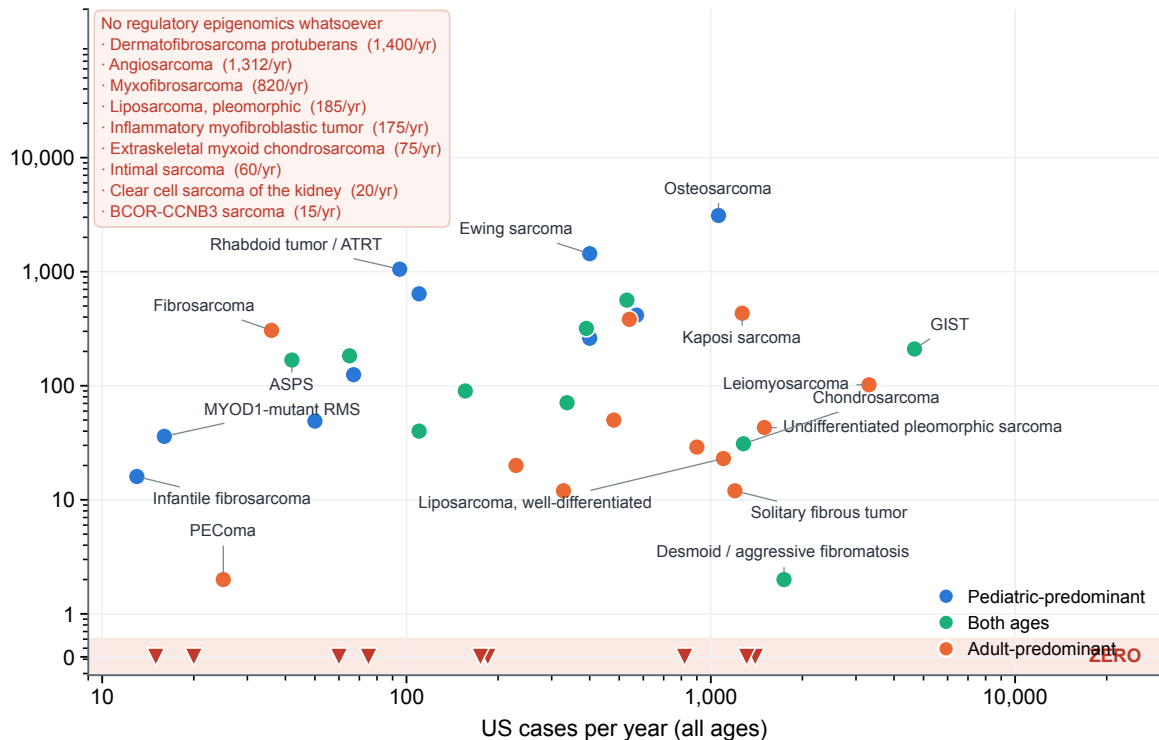


Regulatory epigenomics tracks research history, not disease burden

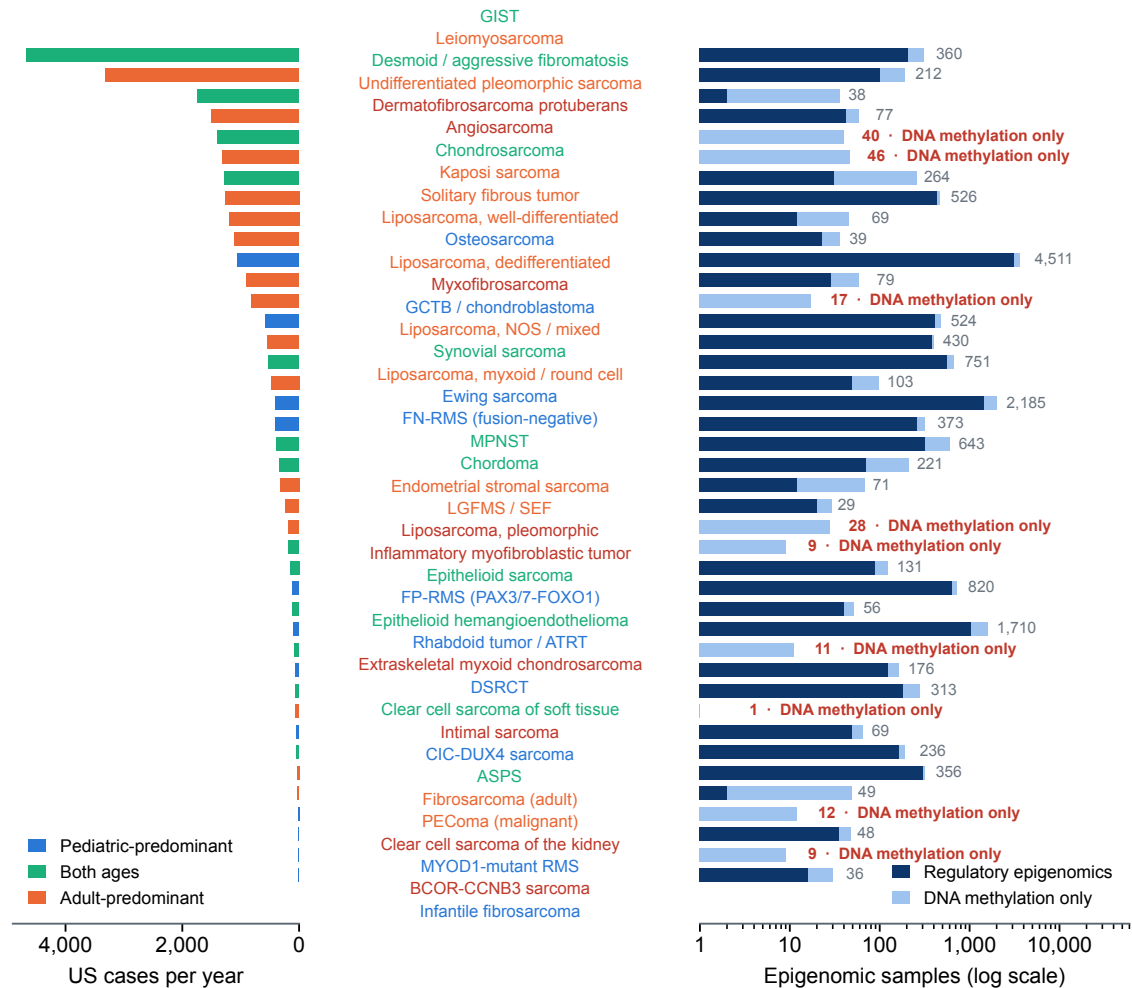
Regulatory epigenomic samples in public archives
(ChIP-seq / CUT&RUN / ATAC / DNase / Hi-C / HiChIP)

No regulatory epigenomics whatsoever

- Dermatofibrosarcoma protuberans (1,400/yr)
- Angiosarcoma (1,312/yr)
- Myxofibrosarcoma (820/yr)
- Liposarcoma, pleomorphic (185/yr)
- Inflammatory myofibroblastic tumor (175/yr)
- Extraskeletal myxoid chondrosarcoma (75/yr)
- Intimal sarcoma (60/yr)
- Clear cell sarcoma of the kidney (20/yr)
- BCOR-CCNB3 sarcoma (15/yr)

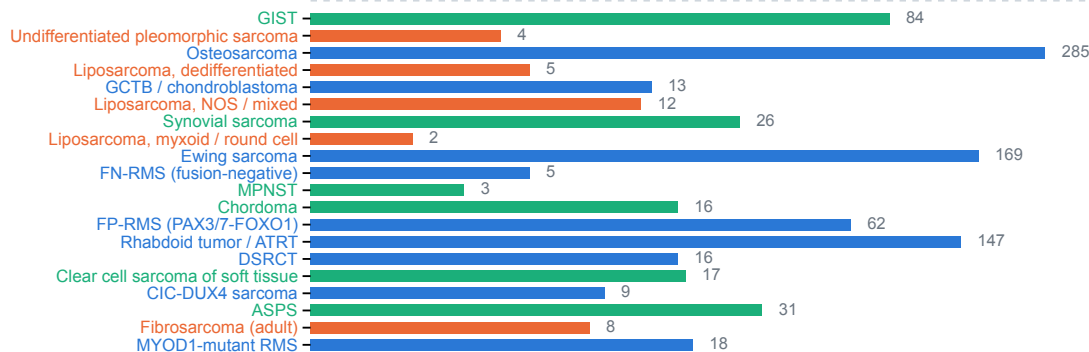


Nine sarcoma entities with a published incidence rate — two of them above 1,300 US cases a year — have DNA methylation profiling and nothing else Data, split by what it can answer



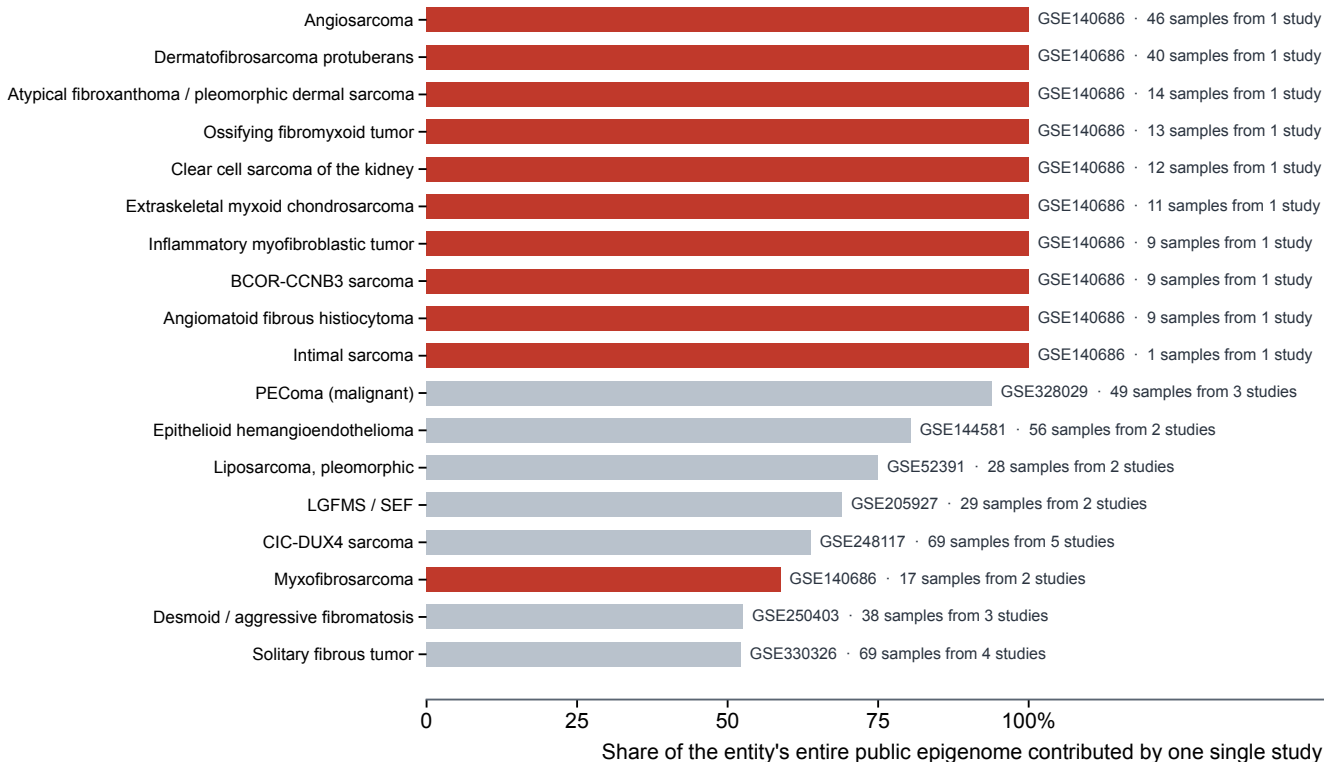
25 of 45 sarcoma entities have never had a single patient's tumor profiled for active chromatin, chromatin accessibility or 3D genome architecture

Leiomyosarcoma	✗	none, against ~3,310 US cases per year
Desmoid / aggressive fibromatosis	✗	none, against ~1,740 US cases per year
Dermatofibrosarcoma protuberans	✗	none, against ~1,400 US cases per year
Angiosarcoma	✗	none, against ~1,312 US cases per year
Chondrosarcoma	✗	none, against ~1,280 US cases per year
Kaposi sarcoma	✗	none, against ~1,268 US cases per year
Solitary fibrous tumor	✗	none, against ~1,200 US cases per year
Liposarcoma, well-differentiated	✗	none, against ~1,100 US cases per year
Myxofibrosarcoma	✗	none, against ~820 US cases per year
Endometrial stromal sarcoma	✗	none, against ~328 US cases per year
LGFMS / SEF	✗	none, against ~229 US cases per year
Liposarcoma, pleomorphic	✗	none, against ~185 US cases per year
Inflammatory myofibroblastic tumor	✗	none, against ~175 US cases per year
Epithelioid sarcoma	✗	none, against ~156 US cases per year
Epithelioid hemangioendothelioma	✗	none, against ~110 US cases per year
Extraskeletal myxoid chondrosarcoma	✗	none, against ~75 US cases per year
Intimal sarcoma	✗	none, against ~60 US cases per year
PEComa (malignant)	✗	none, against ~25 US cases per year
Clear cell sarcoma of the kidney	✗	none, against ~20 US cases per year
BCOR-CCNB3 sarcoma	✗	none, against ~15 US cases per year
Infantile fibrosarcoma	✗	none, against ~13 US cases per year
Myoepithelial carcinoma of soft tissue	✗	none (no population rate published anywhere)
Ossifying fibromyxoid tumor	✗	none (no population rate published anywhere)
Angiomatoid fibrous histiocytoma	✗	none (no population rate published anywhere)
Atypical fibroxanthoma / pleomorphic dermal sarcoma	✗	none (no population rate published anywhere)



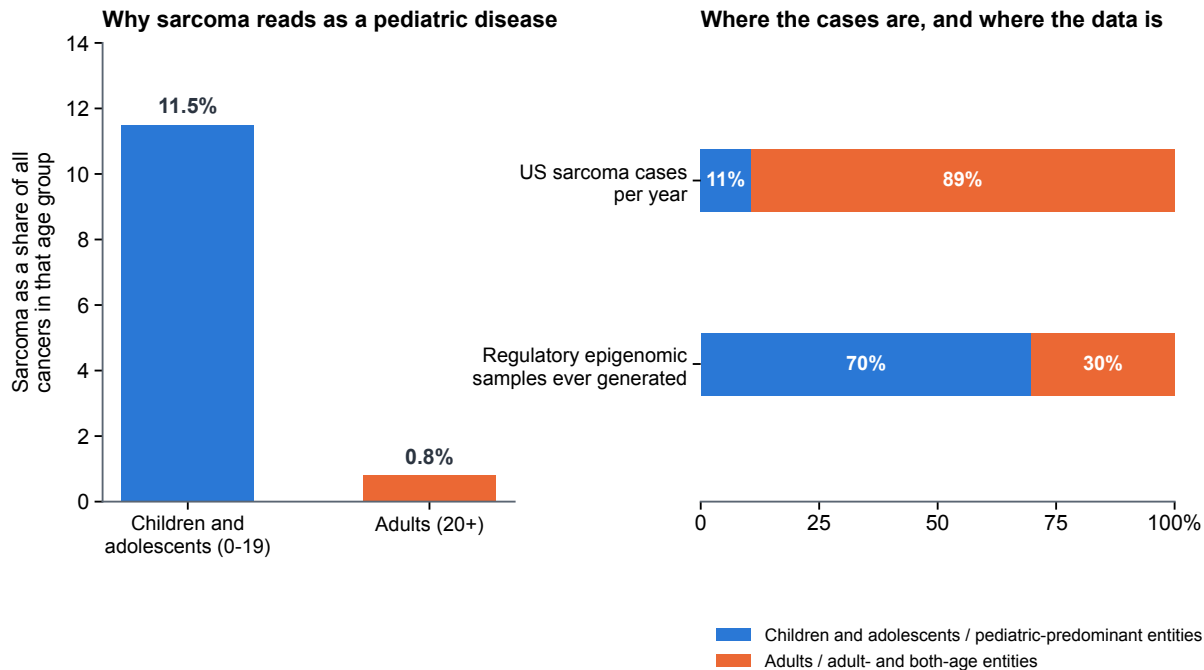
Regulatory epigenomic profiles of PATIENT-DERIVED MATERIAL — tumor tissue, PDX or patient-derived organoid (ChIP-seq / CUT&RUN / ATAC / Hi-C; input and IgG controls excluded)

Most sarcoma entities rest on one deposit — and for many of them that deposit carries no entity label in GEO at all



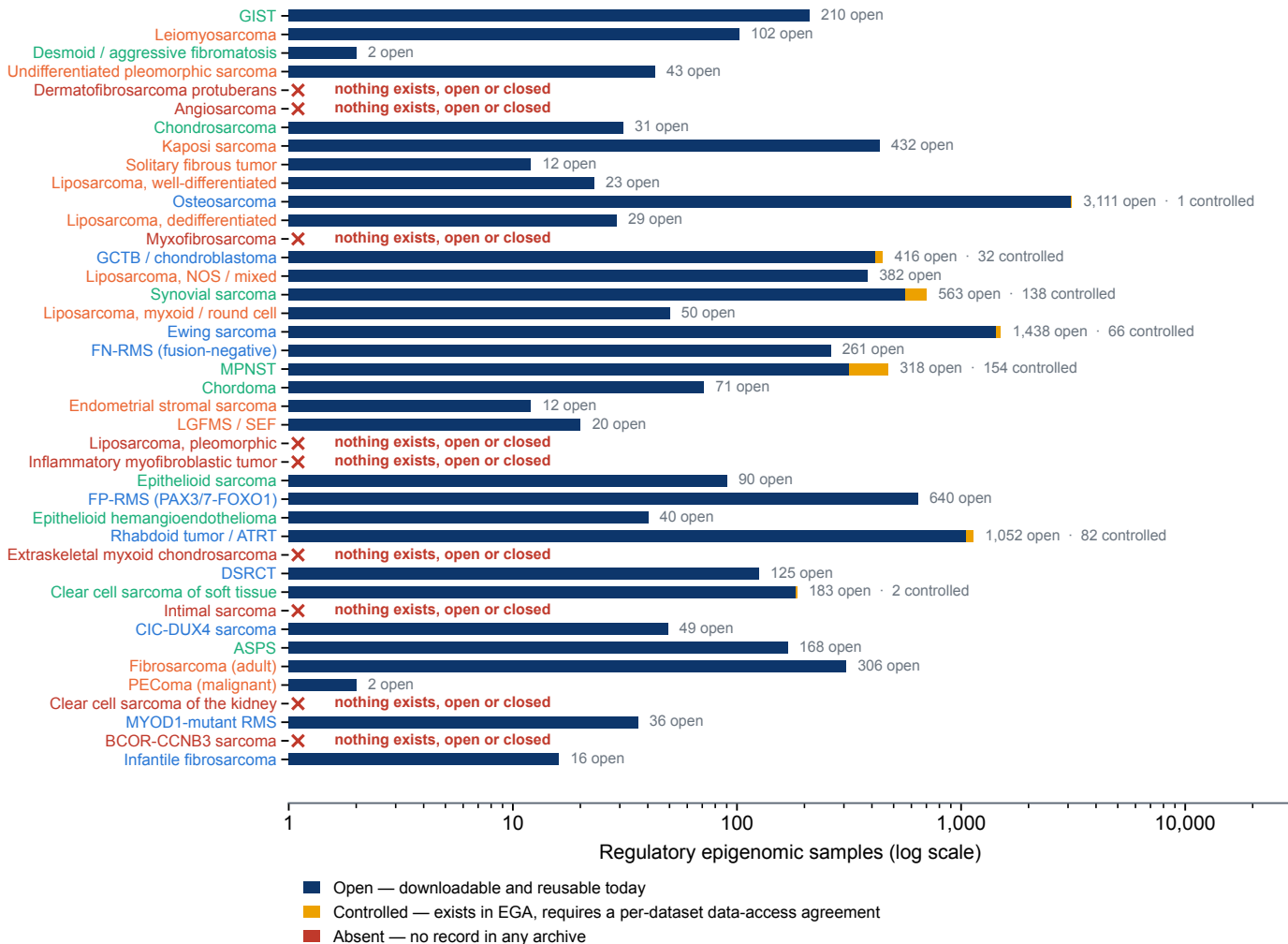
■ GSE140686 — 1,505 arrays deposited with the diagnosis stripped from every sample
■ Some other single study

Sarcoma is 1% of adult cancer and 11% of childhood cancer — and the epigenomic literature followed the 11%



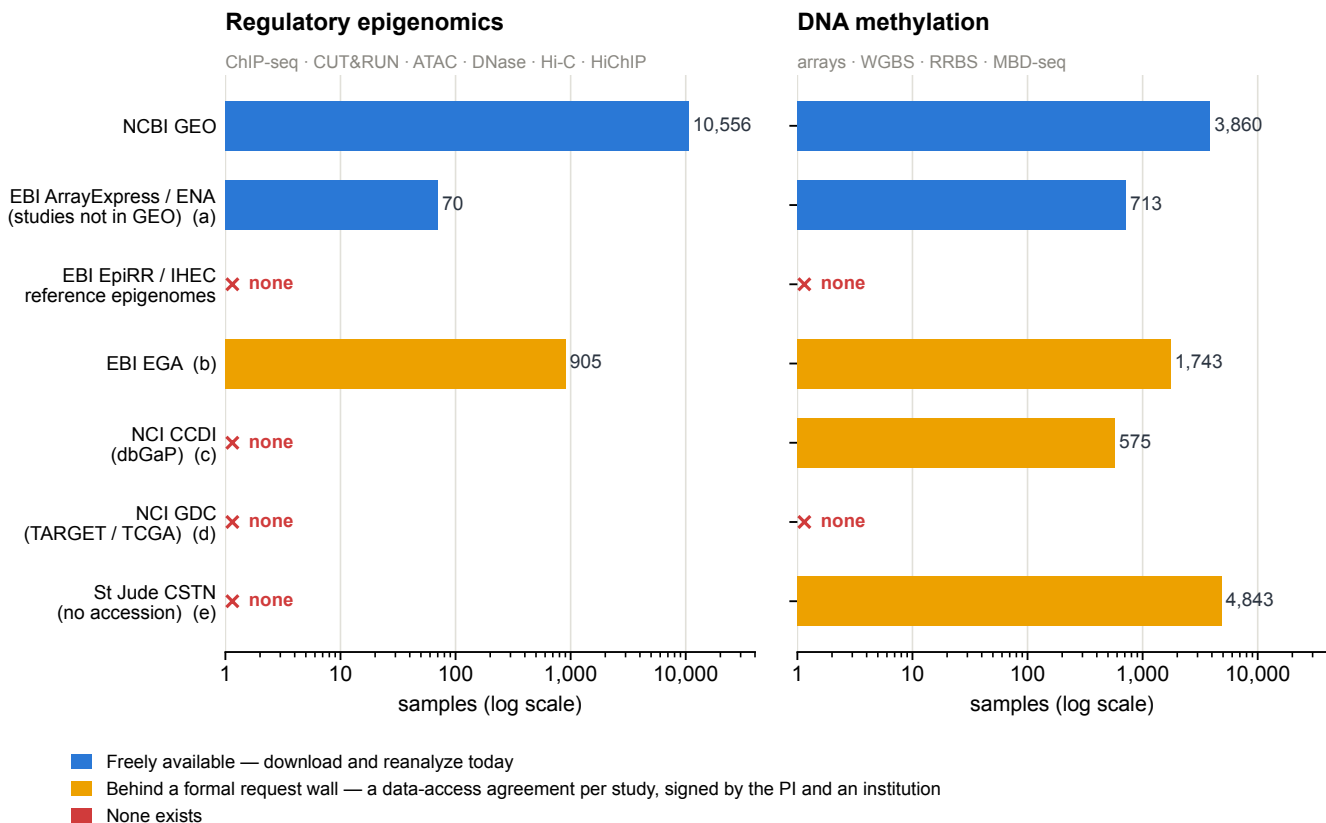
Sources: sarcoma share of all cancers and US case counts, Siegel et al., Cancer Statistics 2026 (PMID 41528114; 4,110 bone + 13,910 soft tissue = 18,020) and SEER 21 age distribution; pediatric share of childhood malignancy, Siegel DA et al., JNCI 2023 (PMID 37433078; ICCG groups VIII + IX = 11.5%). Sample counts from this atlas (T13).

Some of the gap is unmeasured biology; some of it is data that exists and cannot be reused 1,207 regulatory epigenomes sit behind data-access committees — but not one covers the nine entities with nothing at all



A further 430 regulatory epigenomes sit in EGA under controlled access in cohorts whose subtype EGA does not state — RMS (any subtype): 413; Sarcoma, unspecified / mixed cohort: 17. The largest is the St Jude CSTN rhabdomyosarcoma ChIP-seq (EGAD00001004312 + EGAD00001006398), none of which is in GEO. They are excluded from the bars rather than attributed by guesswork.

Regulatory epigenomics for sarcoma is mostly open. DNA methylation is mostly not.
 92% of the 11,531 regulatory samples can be downloaded today, against only 39% of the 11,734 methylation samples.



Unit: samples, except CCDI, which resolves to participants. Duplicates removed; ChIP input and IgG controls excluded from the regulatory counts.

(a) The 23 studies that exist only at EBI, sized from the ENA and BioStudies APIs. E-MTAB-9875 (1,974 assays) is held out — it is the sarcoma methylation classifier, the same material as GSE140686 in GEO.

(b) Includes the St Jude CSTN deposit (469 samples), which is counted here rather than separately because EGA is where it lives. (c) CCDI has no ChIP-seq in its assay vocabulary at all, for any disease.

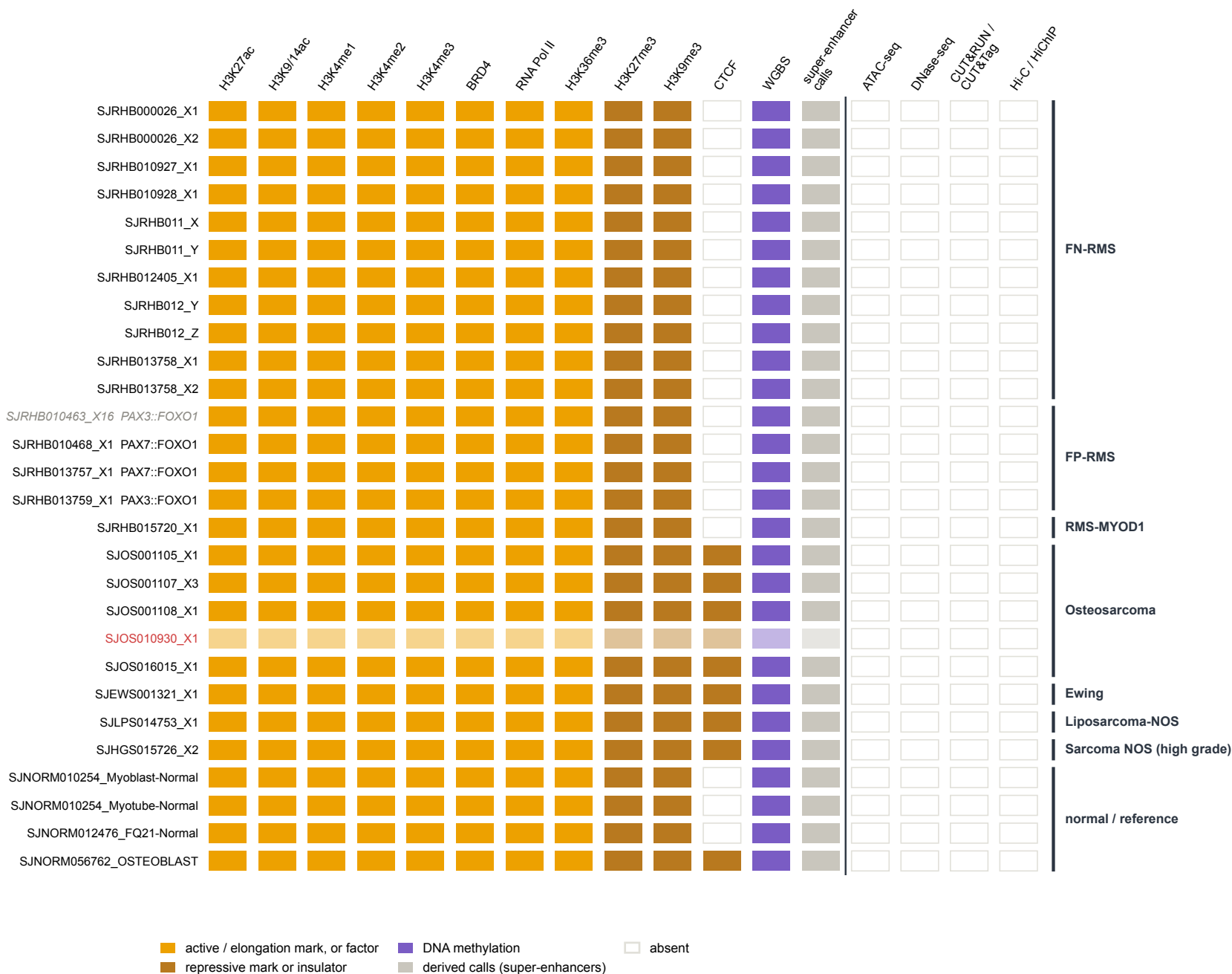
(d) The GDC has no ChIP-seq in its vocabulary either, and its 410 ATAC-seq files belong to 23 TCGA cohorts, none of them SARC. Methylation-array files exist for TARGET and TCGA sarcoma projects but are not sample-resolved here, so they are not counted.

(e) St Jude's COMET methylation project and the St Jude Cloud CSTN dataset. A request route exists for both, so they sit behind the same wall as EGA and dbGaP — but neither has an archive accession, which makes them the weakest case within it.

Also recovered this round: the CSTN ProteinPaint browsers resolve to 278 regulatory tracks on 27 models (T28, F26), with the per-model diagnosis supplied by the St Jude data administrator. That material is already counted once inside the EGA bar, so it is not added again — what it adds is assay-level resolution EGA does not publish.

The densest sarcoma epigenome resource that exists is complete, uniform — and entirely behind a wall.

Every one of the 28 St Jude CSTN models carries the same 10-mark panel plus WGBS and super-enhancer calls (CTCF only in the 2018 study). Right of the rule: what it does not contain at all.



Access: all of it requires a St Jude / EGA data access agreement. None of it is added to any atlas total — the same material is already counted once under EGA (F25). What the enumeration adds is this resolution, which EGA does not publish.

Entity calls are the St Jude data administrator's (email); the fusion partner is from the CSTN portal model table (T17). Both are recorded per track in T28.

Red label: SJOS010930_X1, which the administrator reports as withdrawn from CSTN and no longer available — its tracks are still drawn on the browser page and are shown faded.

Grey italic label: SJRHB010463_X16, which is on the browser page and in the portal as PAX3::FOXO1 but is absent from the administrator's list; the discrepancy is left standing.